

Tswalu Steel Fabrication

Dual-Stack Network Design & Implementation Report

IPv6 Migration · Secure Network Administration · Guest Network Segmentation

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Project Reference	CMPG325-2026-029
Phase	Milestone 2 — Implementation & Testing

Repository

https://github.com/blessedmaake18-bee/Tswalu_Network

Includes the working Cisco Packet Tracer implementation, device configuration exports, full testing evidence, and supporting documentation.

1. Introduction

This report documents the design and implementation of a secure, dual-stack (IPv4/IPv6) network for Tswalu Steel Fabrication, a steel manufacturing company based in Rustenburg. It's the Milestone 2 build-out of the network as designed in the initial phase, implemented and tested in Cisco Packet Tracer on four objectives, reliable connectivity between the five departments, support for native operation of both IPv4 and IPv6, restricting administrative access to authenticated users, and separating the Guest Wi-Fi network completely from the internal network.

It covers what was configured, why, and how each requirement was tested and verified, with supporting evidence for every test performed.

1.1 Project Scope

- A complete, working Cisco Packet Tracer implementation of the network
- Native IPv6 dual-stack addressing and routing, configured and verified
- Testing evidence for connectivity, network security, and traffic segmentation
- Supporting configuration exports and documentation

2. Network Design Overview

The design addresses four objectives: connecting the organisation, introducing IPv6, protecting the infrastructure, and supporting visitors safely. It uses five VLANs routed through a single Layer 3 router (R1) via sub-interfaces on a trunk to the core switch, SW1, on the IPv4 block 172.30.12.0/23 and the IPv6 prefix 2001:DB8:ACAD::/48.

VLAN	Segment / IPv4 Network / IPv6 Prefix
10	Management & Administration — 172.30.12.64/27 — 2001:DB8:ACAD:10::/64
20	Production — 172.30.12.0/26 — 2001:DB8:ACAD:20::/64
30	Finance & HR — 172.30.12.128/28 — 2001:DB8:ACAD:30::/64
40	IT & Services — 172.30.12.144/28 — 2001:DB8:ACAD:40::/64
50	Guest Wi-Fi — 172.30.12.96/27 — 2001:DB8:ACAD:50::/64

3. Implementation Summary

The table below maps each requirement to how it was implemented in Packet Tracer.

Requirement	Implementation
VLAN segmentation & trunking	Five VLANs created on SW1; Fa0/1 trunk to R1 (native VLAN 99); Fa0/2–Fa0/8 as access ports per device
IPv4 addressing & inter-VLAN routing	Router-on-a-stick sub-interfaces (Gi0/0.10–Gi0/0.50) on R1, each with the VLAN's gateway address

IPv6 dual-stack routing	ipv6 unicast-routing enabled; each sub-interface given its /64 gateway address from the design prefix
Secured device administration	Local user authentication (login local), SSH-only remote access (transport input ssh), enable secret, hashed passwords, idle timeouts on R1 and SW1
Guest Wi-Fi isolation	Extended IPv4 ACL and IPv6 ACL applied inbound on the Guest sub-interface (Gi0/0.50), denying guest traffic to all four internal VLANs while permitting everything else

3.1 Host and Management Addressing

The network design specifies each VLAN's gateway and usable address range. The table below records the specific host and management addresses assigned during implementation.

Device	Address(es)
R1 (gateways)	172.30.12.65 / .1 / .129 / .145 / .97 — 2001:DB8:ACAD:{10,20,30,40,50}::1
SW1 (management, VLAN 10 SVI)	172.30.12.66 /27, default-gateway 172.30.12.65
PC-Admin	172.30.12.67 — 2001:DB8:ACAD:10::10
PC-Office	172.30.12.68 — 2001:DB8:ACAD:10::11
PC-Production	172.30.12.2 — 2001:DB8:ACAD:20::10
PC-Finance	172.30.12.130 — 2001:DB8:ACAD:30::10
PC-IT	172.30.12.146 — 2001:DB8:ACAD:40::10
SRV1	172.30.12.147 — 2001:DB8:ACAD:40::11
Laptop-Guest	172.30.12.99 — 2001:DB8:ACAD:50::10

4. IPv6 Dual-Stack Addressing & Routing

One of the key features of this project was to have R1 running simultaneously with both IPv4 and IPv6 and to route IPv6 between all segments. On R1, the global command `ipv6 unicast-routing` was enabled, assigning each of five sub-interfaces an IPv6 address from the design prefix `2001:DB8:ACAD::/48` that is identical to the address of its corresponding VLAN. Each VLAN will automatically be a connected route with no static routing required since each VLAN is directly connected to R1.

4.1 R1 IPv6 Interfaces and Routing Table

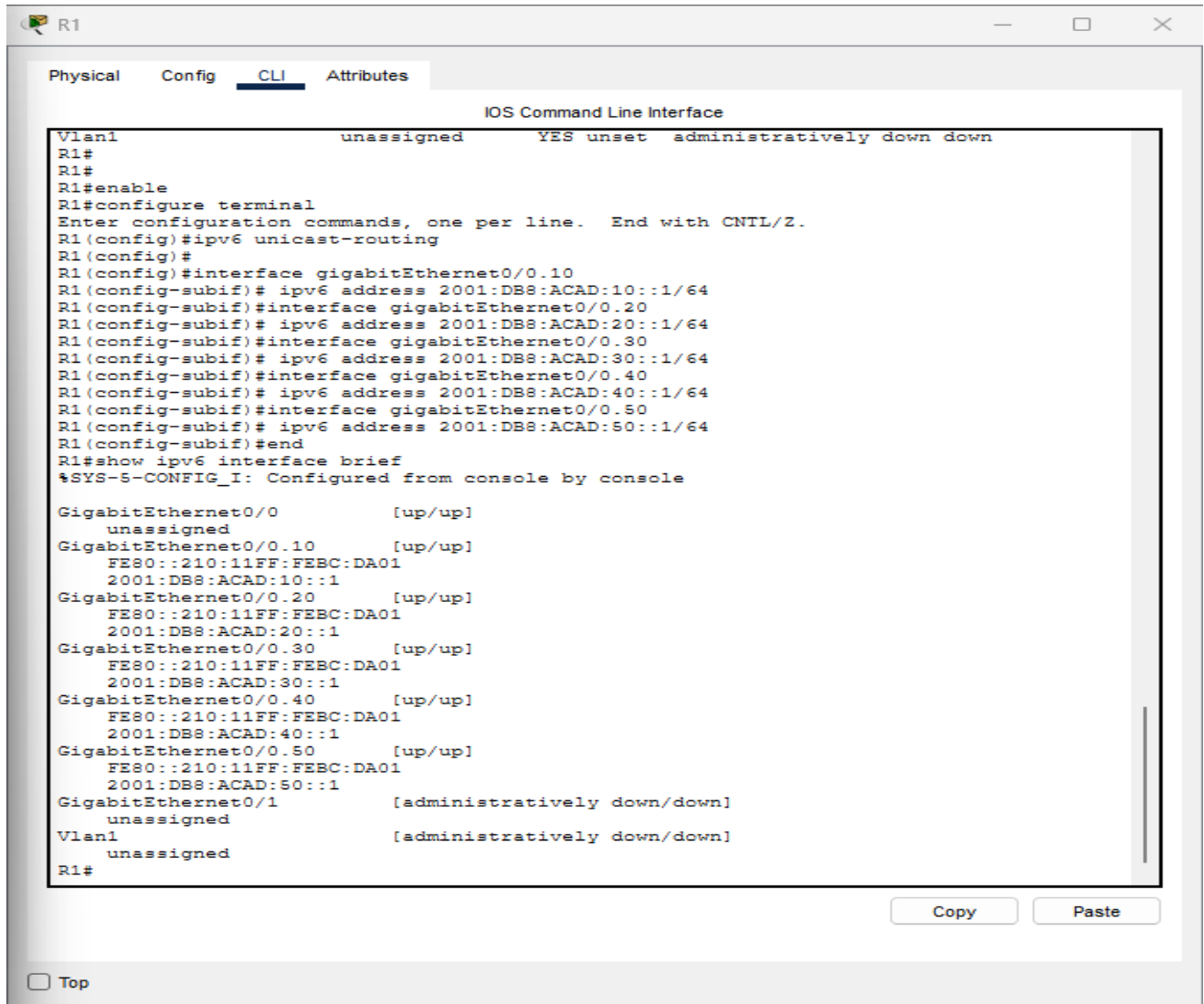


Figure 4.1a — IPv6 unicast-routing enabled; all five sub-interfaces up/up with their /64 addresses

R1

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
R1#
R1#show ipv6 route
IPv6 Routing Table - 11 entries
Codes: C - Connected, L - Local, S - Static, R - RIP, B - BGP
        U - Per-user Static route, M - MIPv6
        I1 - ISIS L1, I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary
        ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
        O - OSPF intra, OI - OSPF inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
        ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
        D - EIGRP, EX - EIGRP external
C   2001:DB8:ACAD:10::/64 [0/0]
    via GigabitEthernet0/0.10, directly connected
L   2001:DB8:ACAD:10::1/128 [0/0]
    via GigabitEthernet0/0.10, receive
C   2001:DB8:ACAD:20::/64 [0/0]
    via GigabitEthernet0/0.20, directly connected
L   2001:DB8:ACAD:20::1/128 [0/0]
    via GigabitEthernet0/0.20, receive
C   2001:DB8:ACAD:30::/64 [0/0]
    via GigabitEthernet0/0.30, directly connected
L   2001:DB8:ACAD:30::1/128 [0/0]
    via GigabitEthernet0/0.30, receive
C   2001:DB8:ACAD:40::/64 [0/0]
    via GigabitEthernet0/0.40, directly connected
L   2001:DB8:ACAD:40::1/128 [0/0]
    via GigabitEthernet0/0.40, receive
C   2001:DB8:ACAD:50::/64 [0/0]
    via GigabitEthernet0/0.50, directly connected
L   2001:DB8:ACAD:50::1/128 [0/0]
    via GigabitEthernet0/0.50, receive
L   FF00::/8 [0/0]
    via Null0, receive
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
R1#
```

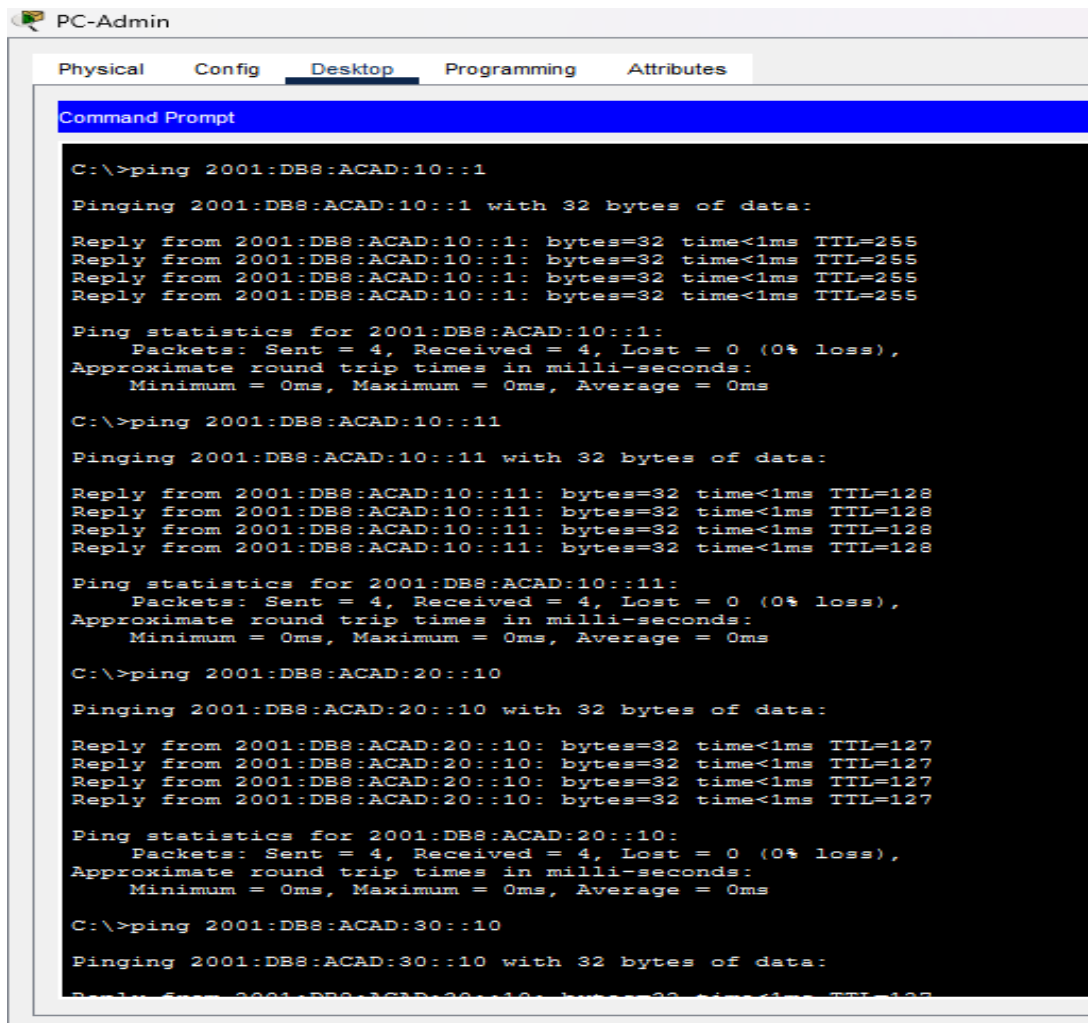
CopyPaste

☐ Top

Figure 4.1b — R1 IPv6 routing table: five connected /64 networks, one per VLAN, each learned automatically

4.2 Verification: IPv6 Ping Tests from PC-Admin

Pings were run from PC-Admin (VLAN 10) to its own gateway, another device in the same VLAN, and devices in three other VLANs. The reply traffic from the same-VLAN side received a TTL of 128, which indicates that no router hop occurred; the reply traffic from the cross-VLAN side received a TTL of 127, which means that the traffic had to be routed through R1.

The image shows a screenshot of a PC-Admin window with the 'Desktop' tab selected. Inside the window is a 'Command Prompt' window with a black background and white text. The text shows four IPv6 ping commands and their results. The first command is 'ping 2001:DB8:ACAD:10::1', which shows four successful replies with a TTL of 255. The second command is 'ping 2001:DB8:ACAD:10::11', showing four successful replies with a TTL of 128. The third command is 'ping 2001:DB8:ACAD:20::10', showing four successful replies with a TTL of 127. The fourth command is 'ping 2001:DB8:ACAD:30::10', which is partially visible at the bottom of the screen.

```
C:\>ping 2001:DB8:ACAD:10::1

Pinging 2001:DB8:ACAD:10::1 with 32 bytes of data:

Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255
Reply from 2001:DB8:ACAD:10::1: bytes=32 time<1ms TTL=255

Ping statistics for 2001:DB8:ACAD:10::1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:ACAD:10::11

Pinging 2001:DB8:ACAD:10::11 with 32 bytes of data:

Reply from 2001:DB8:ACAD:10::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:ACAD:10::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:ACAD:10::11: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:ACAD:10::11: bytes=32 time<1ms TTL=128

Ping statistics for 2001:DB8:ACAD:10::11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:ACAD:20::10

Pinging 2001:DB8:ACAD:20::10 with 32 bytes of data:

Reply from 2001:DB8:ACAD:20::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:20::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:20::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:20::10: bytes=32 time<1ms TTL=127

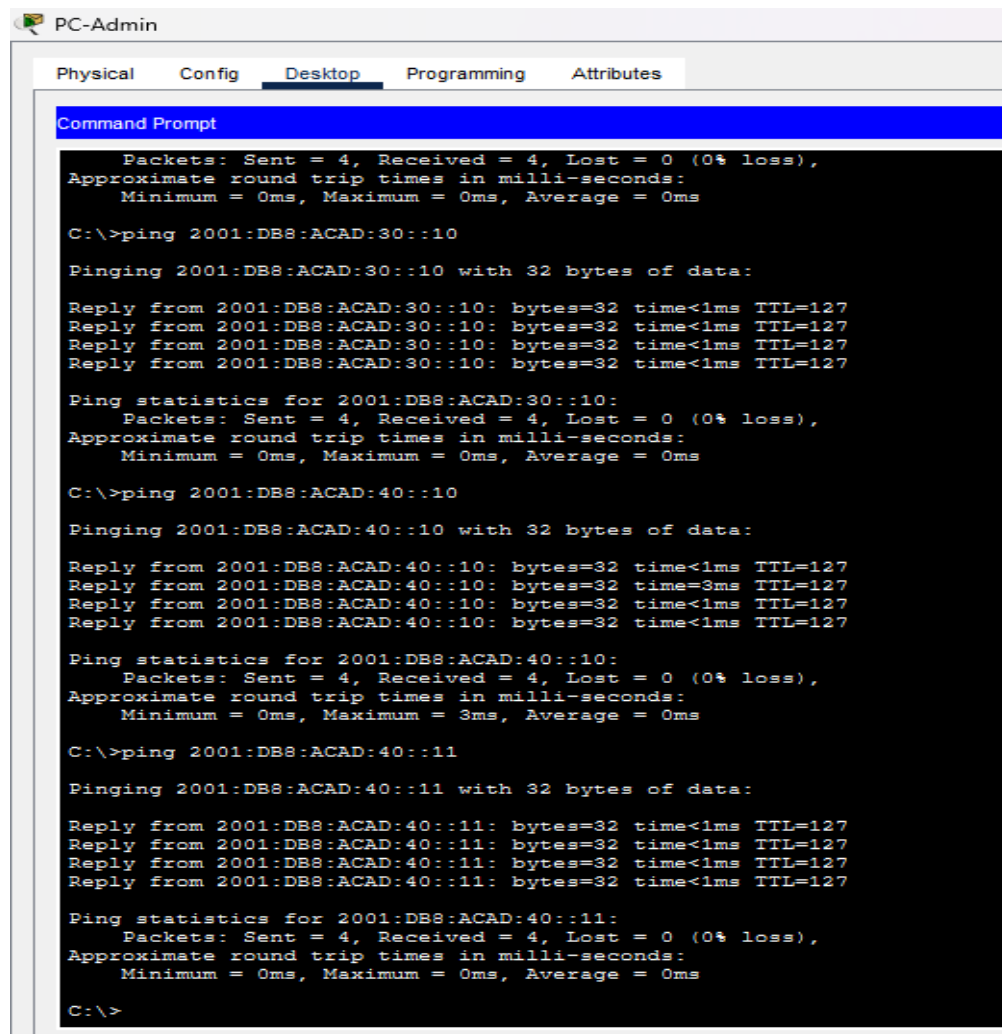
Ping statistics for 2001:DB8:ACAD:20::10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:ACAD:30::10

Pinging 2001:DB8:ACAD:30::10 with 32 bytes of data:

Reply from 2001:DB8:ACAD:30::10: bytes=32 time<1ms TTL=127
```

Figure 4.2a — IPv6 ping to gateway (2001:DB8:ACAD:10::1) and same-VLAN host (::11): 0% loss



```
PC-Admin
Physical  Config  Desktop  Programming  Attributes

Command Prompt

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:ACAD:30::10

Pinging 2001:DB8:ACAD:30::10 with 32 bytes of data:

Reply from 2001:DB8:ACAD:30::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:30::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:30::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:30::10: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:30::10:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 2001:DB8:ACAD:40::10

Pinging 2001:DB8:ACAD:40::10 with 32 bytes of data:

Reply from 2001:DB8:ACAD:40::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:40::10: bytes=32 time=3ms TTL=127
Reply from 2001:DB8:ACAD:40::10: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:40::10: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:40::10:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 2001:DB8:ACAD:40::11

Pinging 2001:DB8:ACAD:40::11 with 32 bytes of data:

Reply from 2001:DB8:ACAD:40::11: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:40::11: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:40::11: bytes=32 time<1ms TTL=127
Reply from 2001:DB8:ACAD:40::11: bytes=32 time<1ms TTL=127

Ping statistics for 2001:DB8:ACAD:40::11:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 4.2b — IPv6 ping to VLAN 30 and VLAN 40 hosts, routed via R1 (TTL 127): 0% loss

Result: IPv6 dual-stack addressing and routing is fully functional and verified across all five VLANs. PASS.

5. Design Constraint: Secured Device Administration

All device administration is to be secured, in line with a previous incident of unauthorised changes in the project brief. The R1 and SW1 had the same configuration: local authentication and SSH-only remote access mode; that is, only users with the appropriate credentials can view or update the configuration.

5.1 Configuration Applied (R1 shown; SW1 identical in structure)

- enable secret — hashes the privileged-mode password
- username netadmin secret — creates a local administrative account
- service password-encryption — obscures any remaining plain-text passwords in the config
- ip domain-name and crypto key generate rsa (2048-bit) — required to support SSH

- ip ssh version 2 — restricts SSH to the more secure version
- line console 0 / line vty 0 4 / line vty 5 15: login local — requires the netadmin login on every console and remote line
- line vty: transport input ssh — blocks Telnet, which sends credentials in clear text
- exec-timeout 5 0 — logs out idle administrative sessions after five minutes

The screenshot shows a Cisco IOS CLI window titled "R1" with tabs for Physical, Config, CLI, and Attributes. The CLI tab is active, displaying the "IOS Command Line Interface". The user is in configuration mode, and the following commands and outputs are shown:

```

R1(config)#username netadmin secret BlessMaake18
R1(config)#end
R1#show running-config | include enable secret|username
%SYS-5-CONFIG_I: Configured from console by console

enable secret 5 $1$mERr$ob7zFMGdia3iEkBkfDGhV1
username netadmin secret 5 $1$mERr$tQlwaAzUQnCsE0LrhZu8t.
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#ip domain-name tswalu.local
R1(config)#no ip domain-lookup
R1(config)#service password-encryption
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
R1#show running-config | include domain|encryption
service password-encryption
no ip domain-lookup
ip domain-name tswalu.local
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#crypto key generate rsa
The name for the keys will be: R1.tswalu.local
Choose the size of the key modulus in the range of 360 to 4096 for your
General Purpose Keys. Choosing a key modulus greater than 512 may take
a few minutes.

How many bits in the modulus [512]: 2048
% Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2
*Mar 1 1:16:40.250: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#end
R1#show ip ssh
%SYS-5-CONFIG_I: Configured from console by console

SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#
  
```

At the bottom of the window, there are "Copy" and "Paste" buttons.

Figure 5.1a — R1: enable secret and netadmin user hashed; domain, SSH keys and SSH v2 enabled

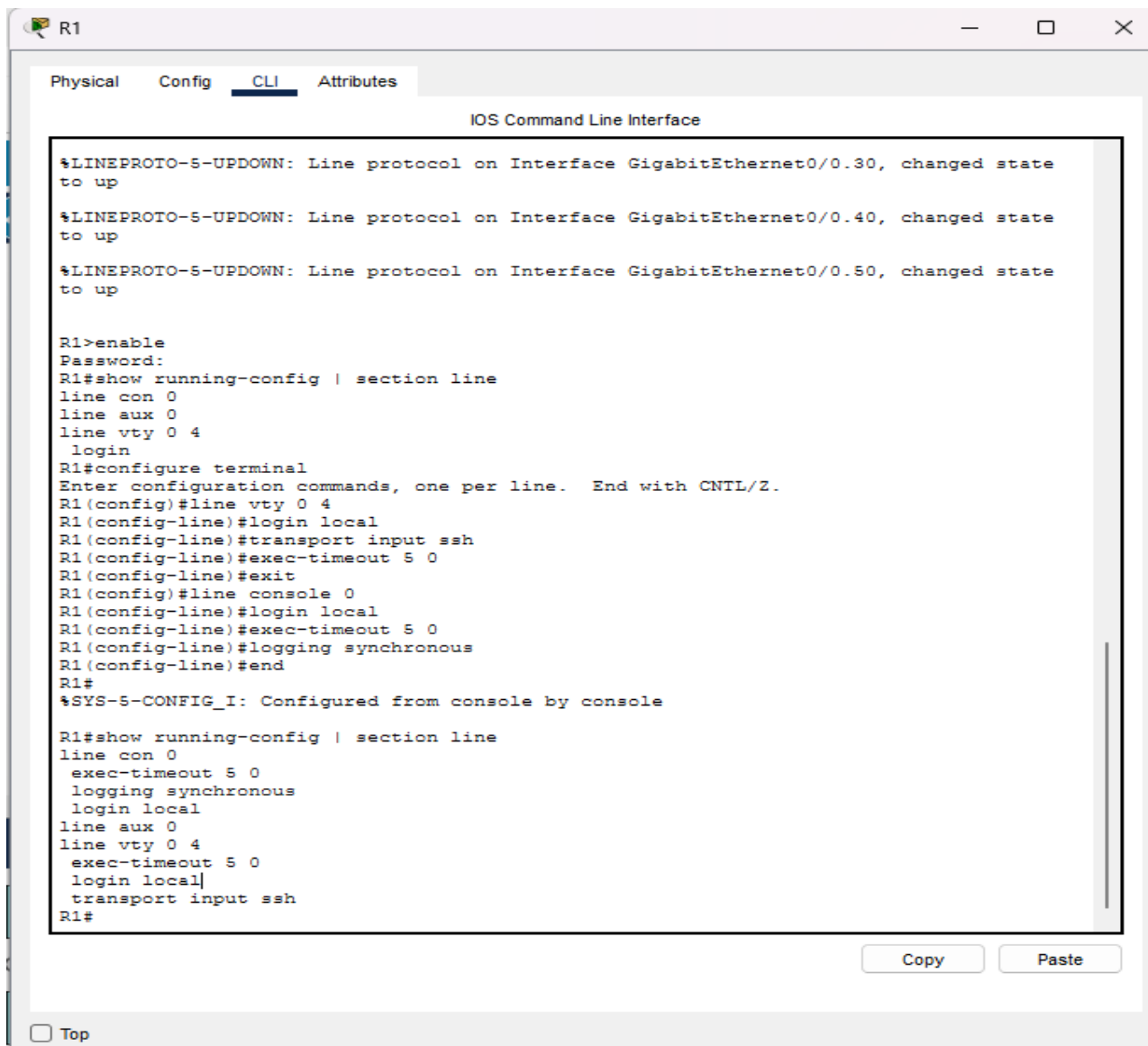


Figure 5.1b — R1: login local and transport input ssh applied to console and VTY lines

5.2 Verification: SSH Login, R1

SSH access to R1 (172.30.12.65) was tested from PC-Admin with both a correct and an incorrect netadmin password, to confirm that only authorised credentials are accepted.

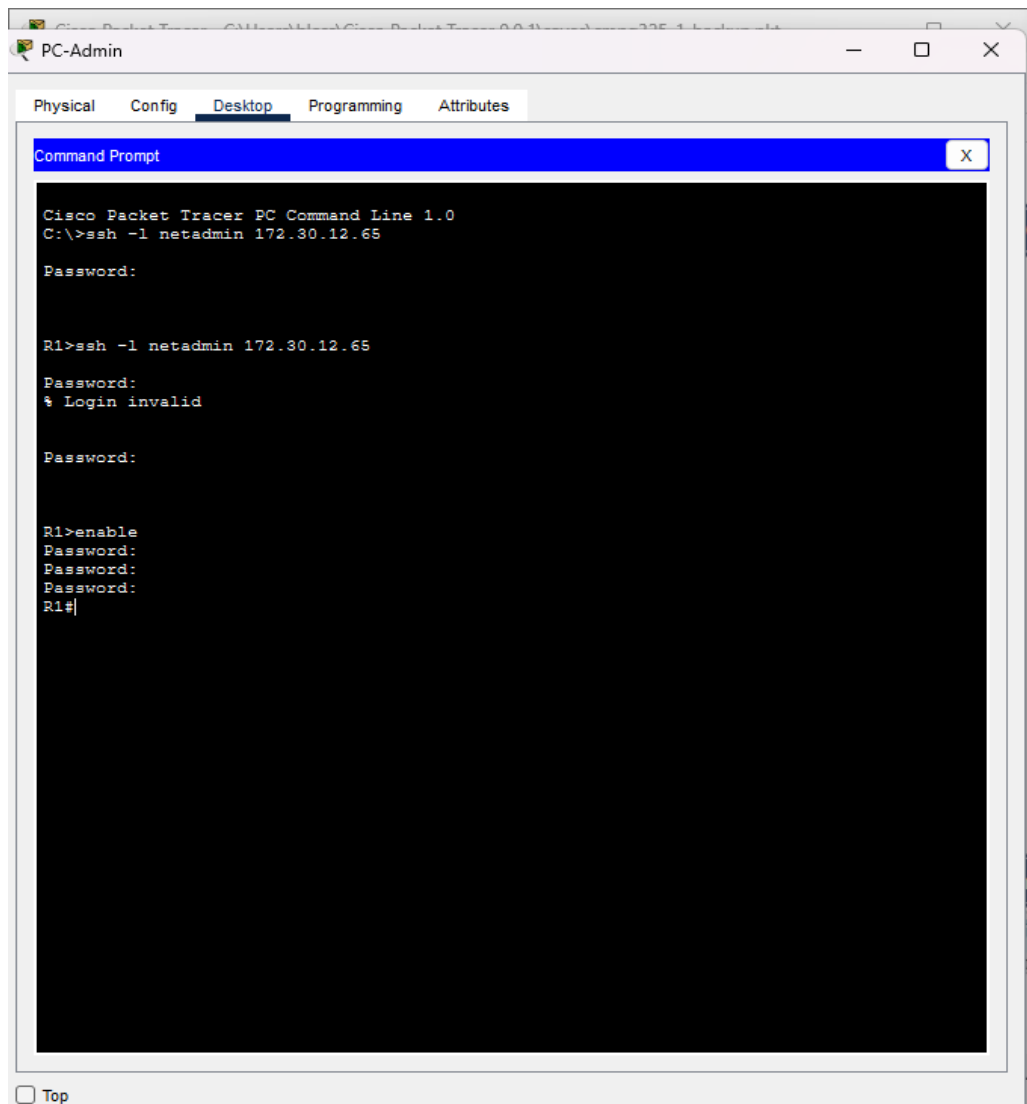


Figure 5.2 — R1: an incorrect password is rejected (% Login invalid); the correct credentials reach R1> and then R1# after enable

5.3 Verification: SSH Login, SW1

SW1 was given a management address on the VLAN 10 SVI (172.30.12.66) and secured identically to R1. SSH access was tested the same way.

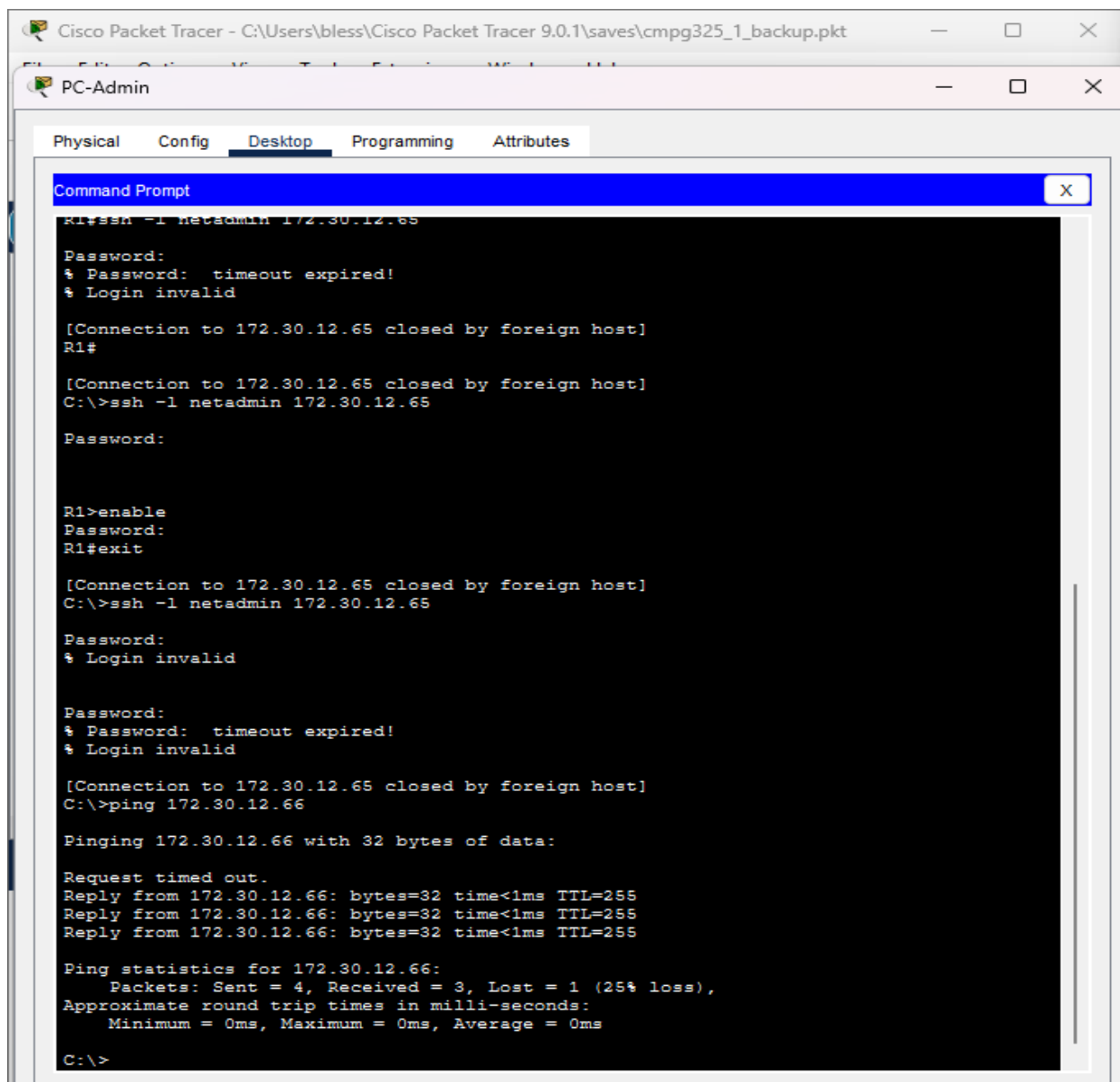


Figure 5.3a — PC-Admin reaches SW1's management address (172.30.12.66) before testing SSH

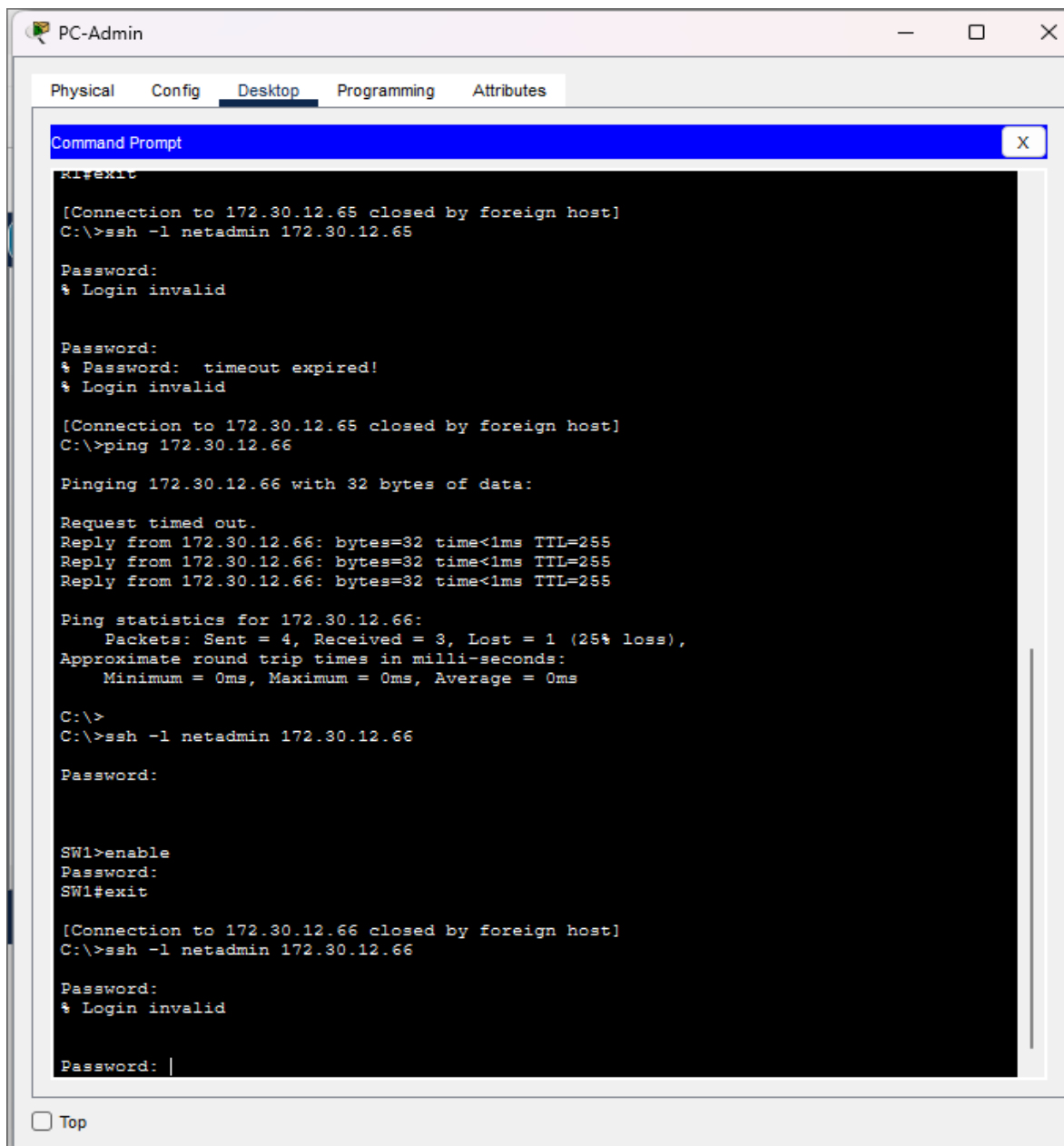


Figure 5.3b — SW1: SSH login rejected with an incorrect password (% Login invalid)

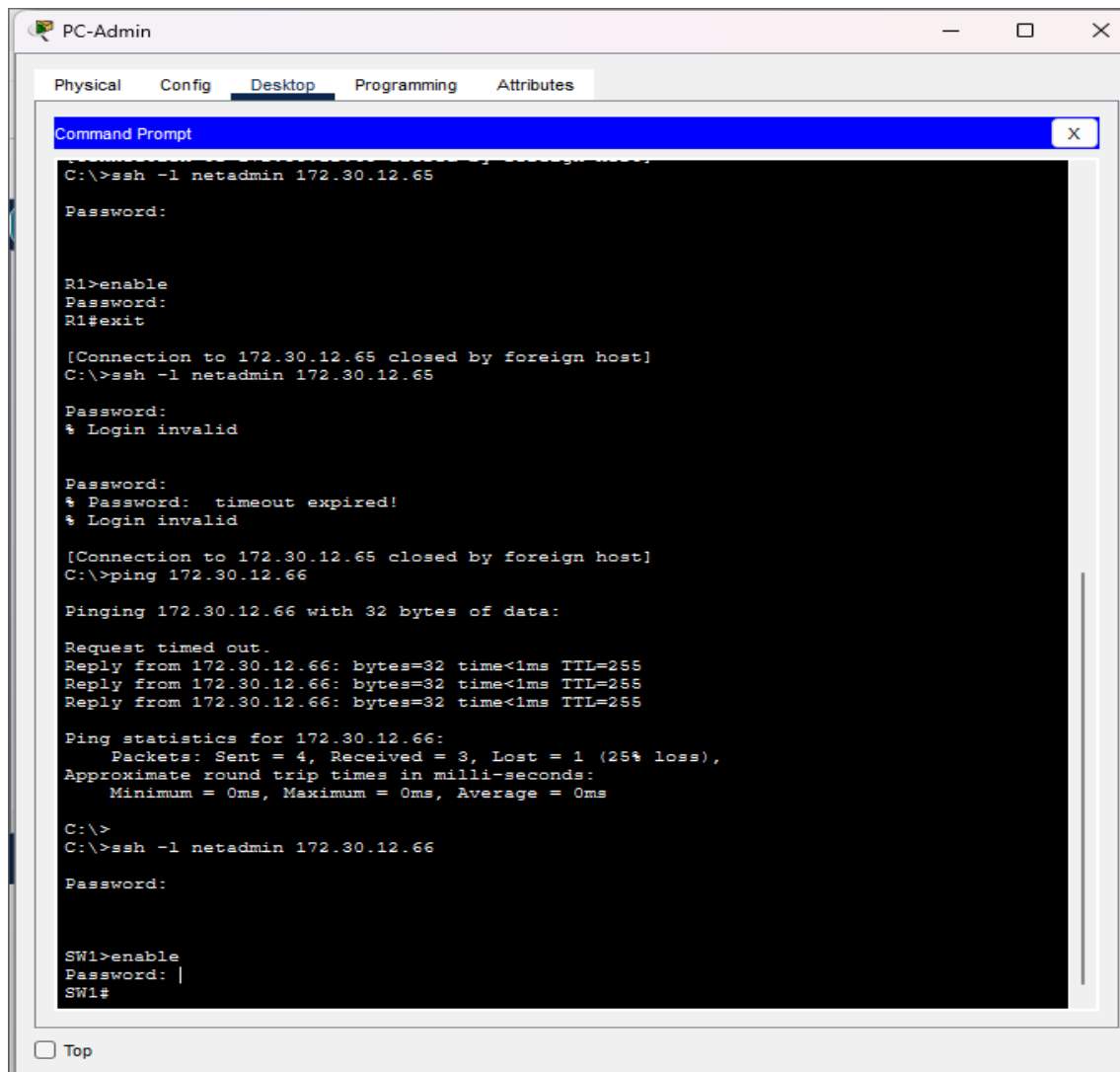


Figure 5.3c — SW1: SSH login succeeds with the correct netadmin credentials, reaching SW1#

Result: both R1 and SW1 require authenticated SSH login; unauthorised attempts are rejected; Telnet is disabled. PASS.

6. Guest Wi-Fi Network Isolation

The client needed to provide Wi-Fi for their visitors and it should be completely separate from their own organisation's resources. VLAN separation isn't enough; an access control list (ACL) was needed in both IPv4 and IPv6 to explicitly deny guest traffic from reaching the four internal VLANs while allowing general traffic to flow.

6.1 IPv4 Guest Isolation ACL

An extended IPv4 ACL (GUEST-ISOLATION) was applied inbound on R1's Guest sub-interface (Gi0/0.50), denying traffic from the Guest subnet to each internal VLAN subnet and permitting everything else.

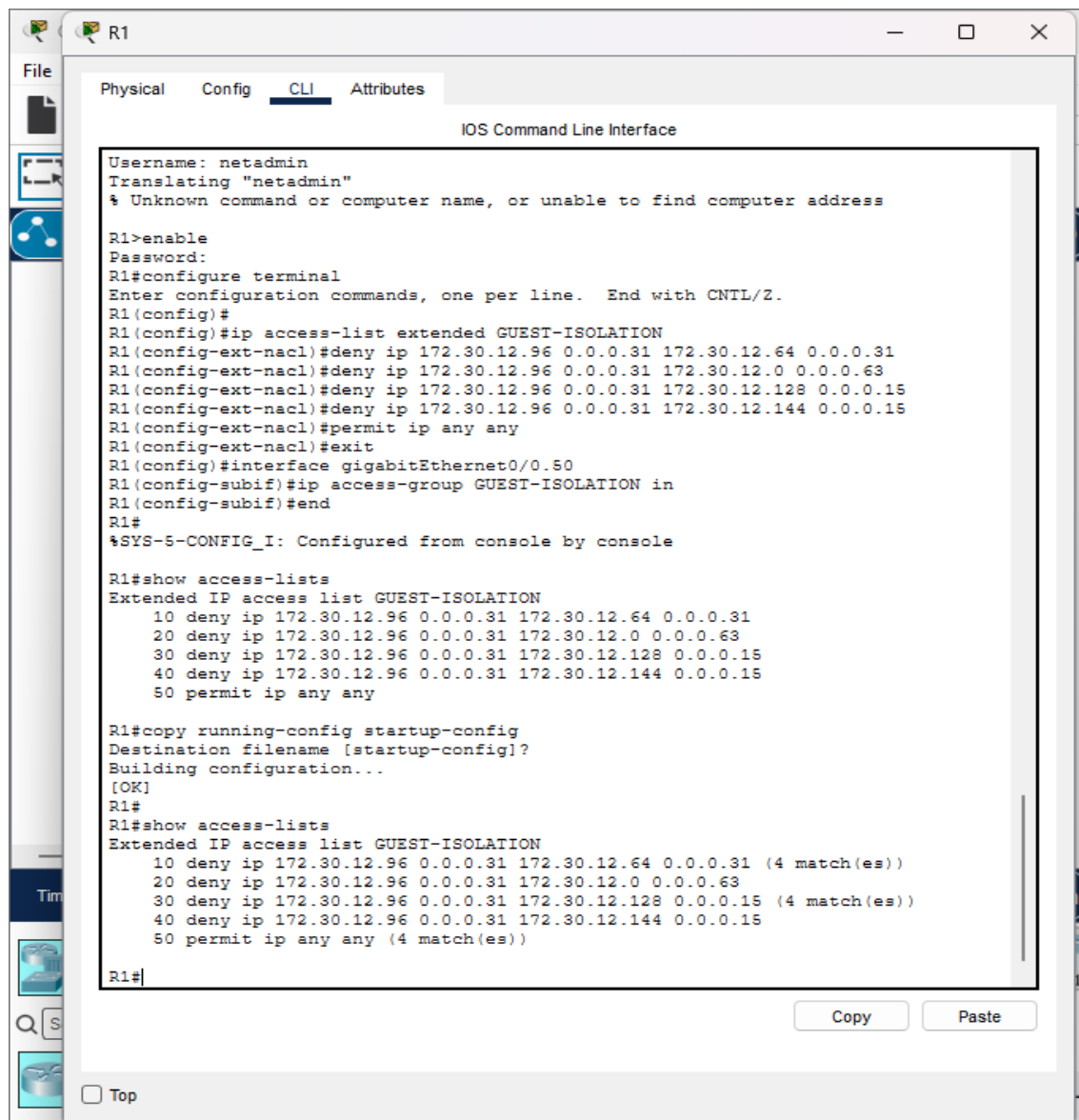


Figure 6.1 — GUEST-ISOLATION ACL on R1, with match counters confirming it is actively filtering traffic

Verification from Laptop-Guest: the Guest gateway (172.30.12.97) replies normally, while VLAN 10 (172.30.12.65) and VLAN 30 (172.30.12.130) are blocked with "Destination host unreachable", returned by R1 itself.

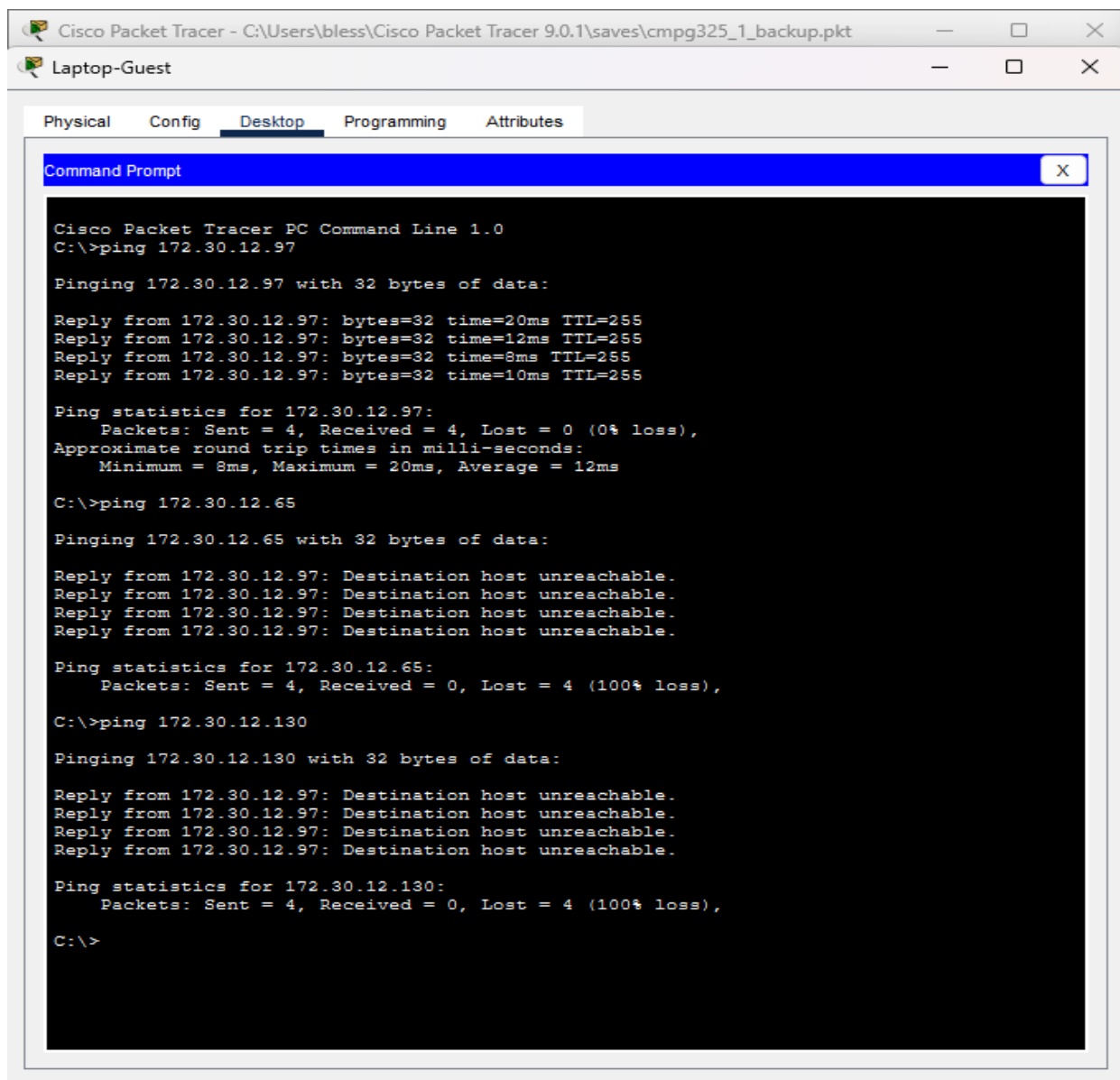


Figure 6.1b — Laptop-Guest: gateway reachable; internal VLANs blocked by the ACL (100% loss)

6.2 IPv6 Guest Isolation ACL

Because the network is dual-stack, an equivalent IPv6 ACL (GUEST-ISOLATION-V6) was required — without it, a guest device could bypass the IPv4 block simply by using IPv6.

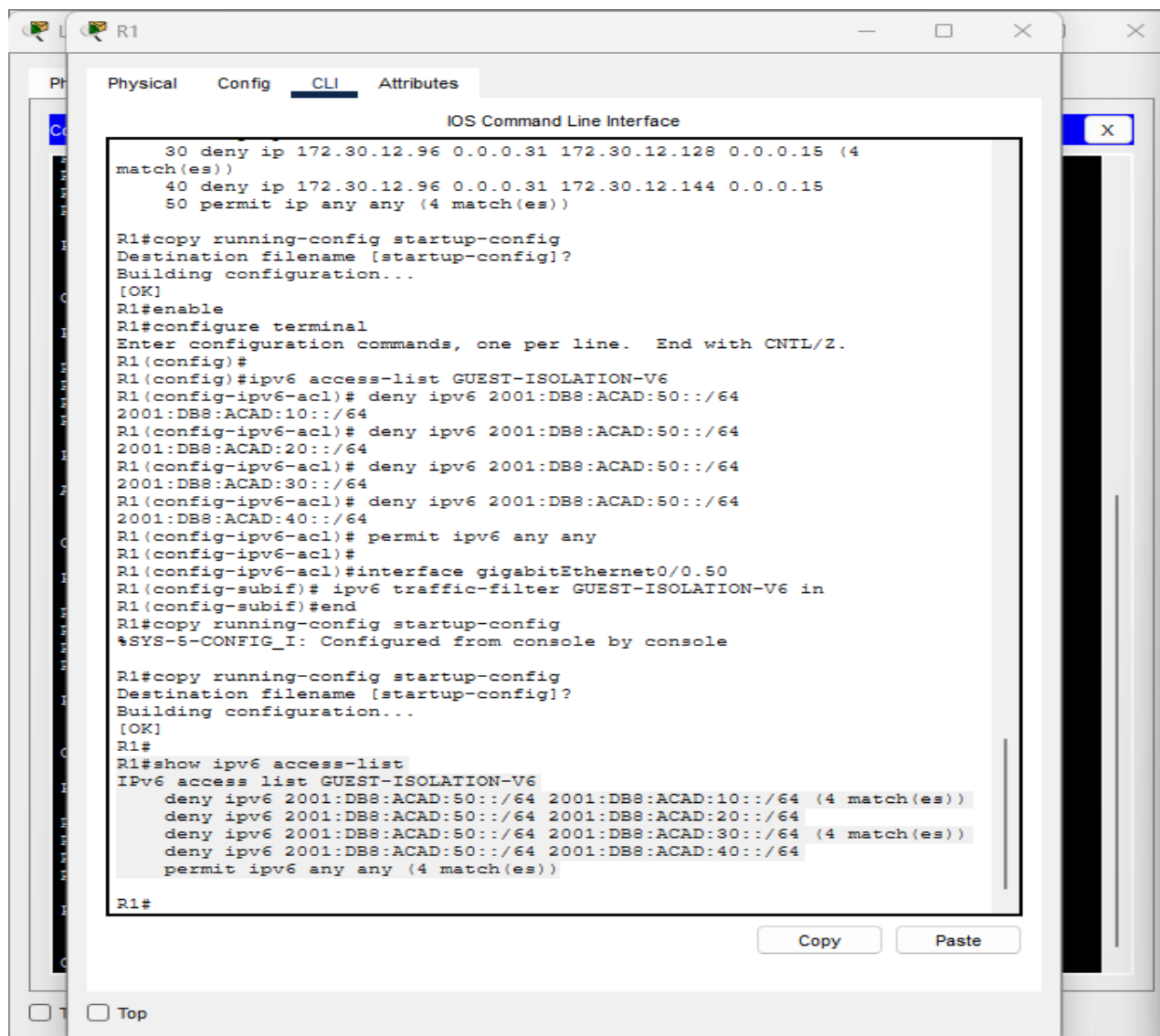


Figure 6.2a — GUEST-ISOLATION-V6 ACL on R1, with match counters confirming it is active

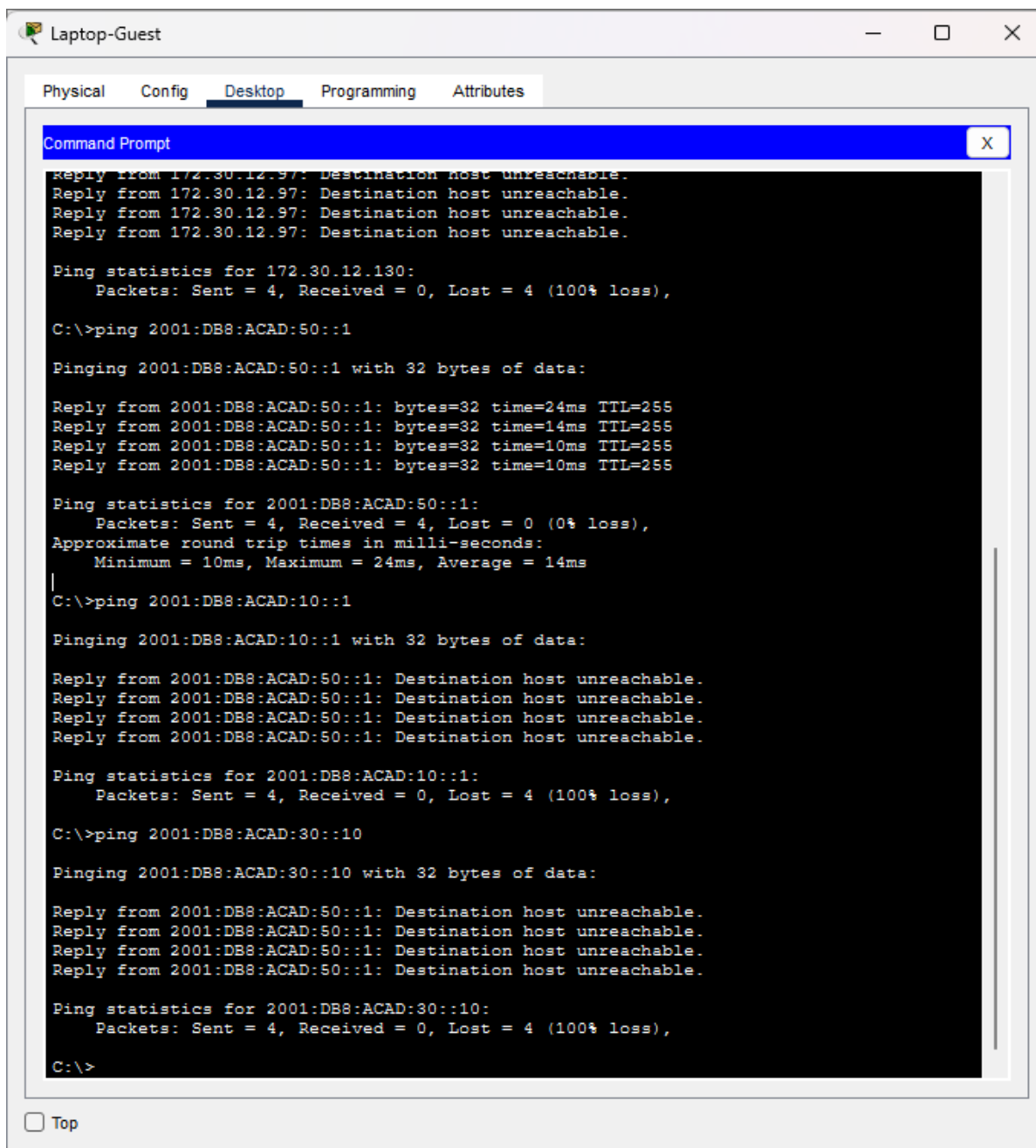


Figure 6.2b — Laptop-Guest: IPv6 gateway reachable; internal VLANs blocked (100% loss)

6.3 Guest Isolation Confirmed via Web Browser

As an additional, visual confirmation, SRV1 (172.30.12.147, VLAN 40) was browsed over HTTP from an internal PC and from Laptop-Guest.

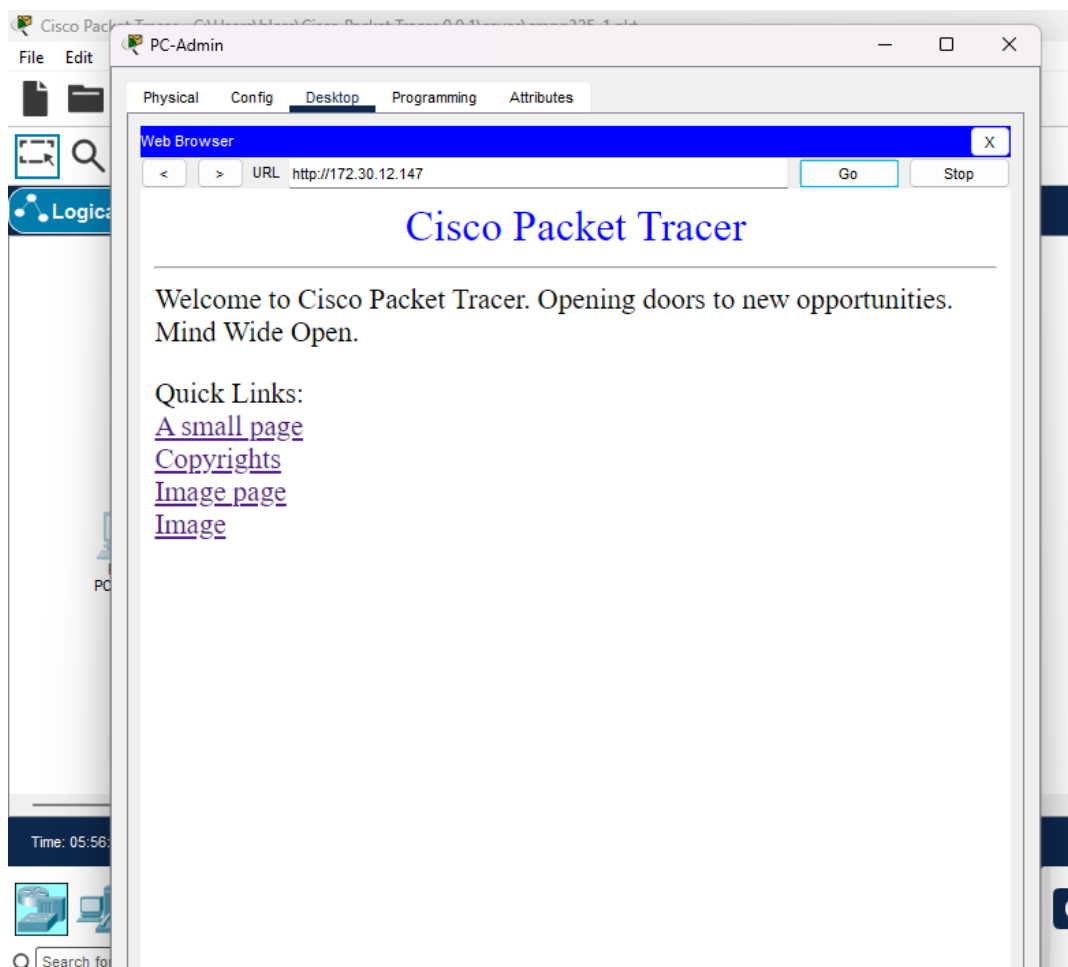


Figure 6.3a — PC-Admin (internal VLAN): SRV1's web page loads successfully

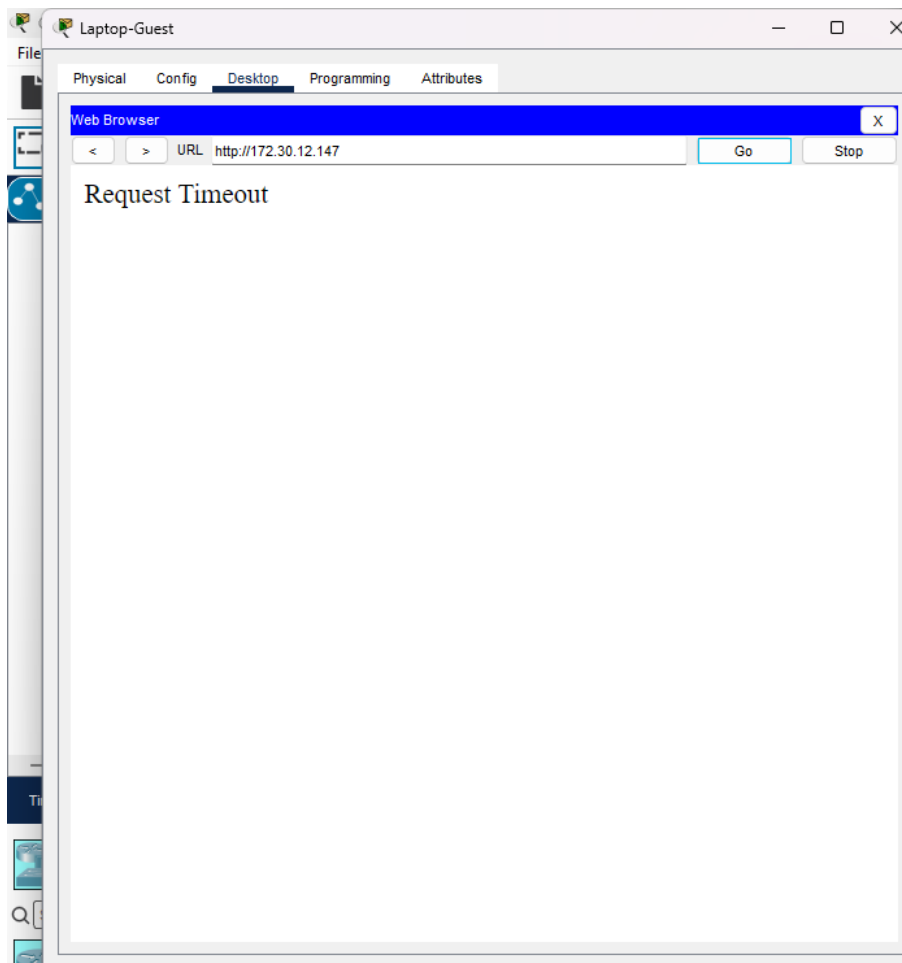


Figure 6.3b — Laptop-Guest: the same request times out, confirming isolation

Result: Guest Wi-Fi (VLAN 50) has general connectivity but is blocked from every internal VLAN, over both IPv4 and IPv6. CR3 satisfied. PASS.

7. Supporting Evidence: IPv4 Base Connectivity

Before layering IPv6, security and guest isolation on top, the underlying IPv4 VLAN segmentation and inter-VLAN routing were configured and tested first.

R1

Physical
Config
CLI
Attributes

IOS Command Line Interface

```

R1(config-subif)# encapsulation dot1q 50
R1(config-subif)# ip address 172.30.12.145 255.255.255.240
R1(config-subif)#
R1(config-subif)#interface gigabitEthernet0/0.50
R1(config-subif)# description Guest-WiFi
R1(config-subif)# encapsulation dot1q 50
R1(config-subif)# ip address 172.30.12.97 255.255.255.224
R1(config-subif)#end
R1#show ip interface brief
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.30, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.40, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.40, changed state to up

%LINK-3-UPDOWN: Interface GigabitEthernet0/0.50, changed state to down

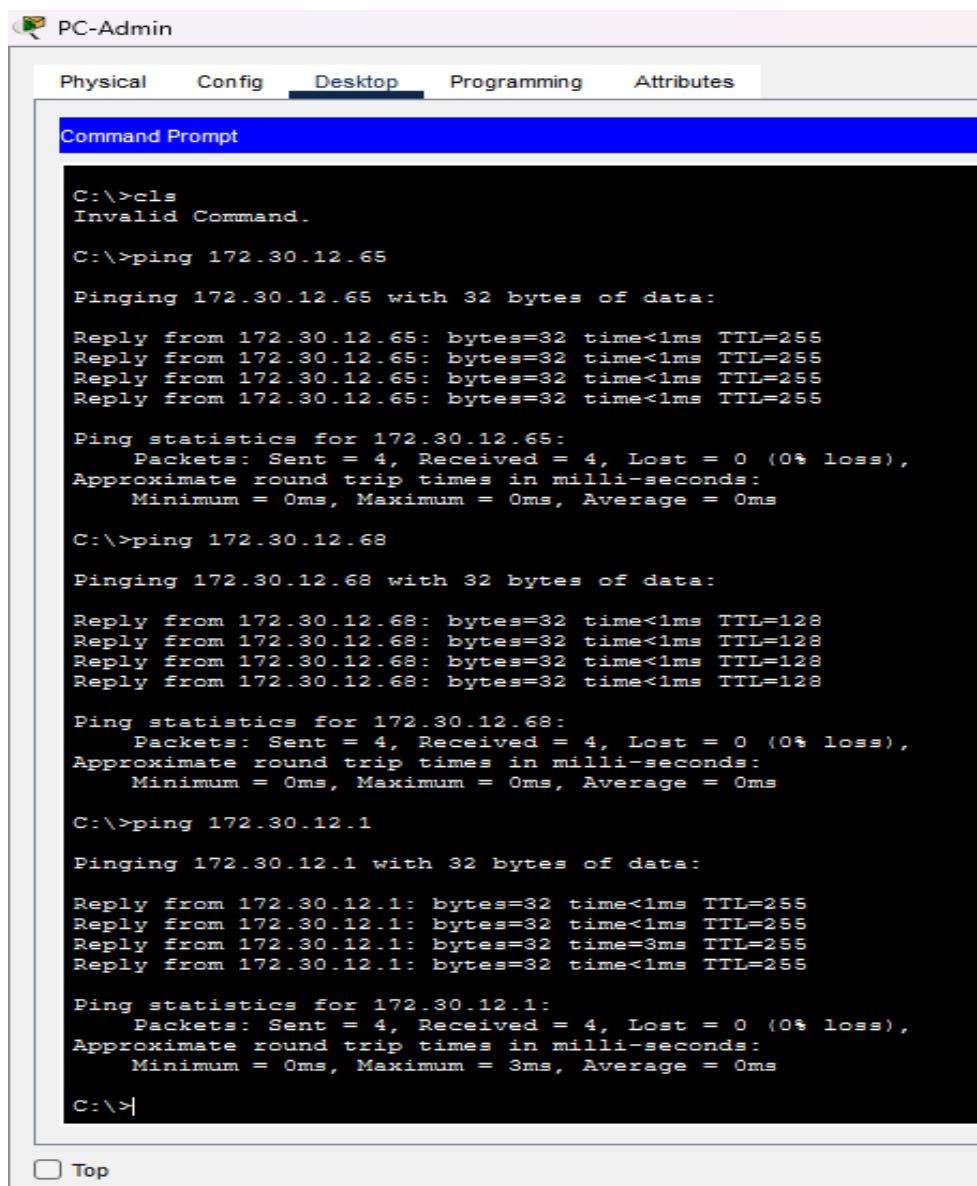
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.50, changed state to up

%SYS-5-CONFIG_I: Configured from console by console

Interface                IP-Address      OK? Method Status          Protocol
GigabitEthernet0/0       unassigned      YES unset    up               up
GigabitEthernet0/0.10    172.30.12.65    YES manual  up               up
GigabitEthernet0/0.20    172.30.12.1     YES manual  up               up
GigabitEthernet0/0.30    172.30.12.129   YES manual  up               up
GigabitEthernet0/0.40    172.30.12.145   YES manual  up               up
GigabitEthernet0/0.50    172.30.12.97    YES manual  up               up
GigabitEthernet0/1       unassigned      YES unset    administratively down down
Vlan1                    unassigned      YES unset    administratively down down
R1#

```

Figure 7.1 — R1: all five IPv4 sub-interfaces configured and up/up



The image shows a screenshot of a PC-Admin application window. The window has a title bar with a small icon and the text "PC-Admin". Below the title bar is a tabbed interface with five tabs: "Physical", "Config", "Desktop", "Programming", and "Attributes". The "Desktop" tab is currently selected. Inside the "Desktop" tab, there is a "Command Prompt" section with a black background and white text. The text shows the following commands and their outputs:

```
C:\>cls
Invalid Command.

C:\>ping 172.30.12.65

Pinging 172.30.12.65 with 32 bytes of data:

Reply from 172.30.12.65: bytes=32 time<1ms TTL=255
Reply from 172.30.12.65: bytes=32 time<1ms TTL=255
Reply from 172.30.12.65: bytes=32 time<1ms TTL=255
Reply from 172.30.12.65: bytes=32 time<1ms TTL=255

Ping statistics for 172.30.12.65:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.12.68

Pinging 172.30.12.68 with 32 bytes of data:

Reply from 172.30.12.68: bytes=32 time<1ms TTL=128
Reply from 172.30.12.68: bytes=32 time<1ms TTL=128
Reply from 172.30.12.68: bytes=32 time<1ms TTL=128
Reply from 172.30.12.68: bytes=32 time<1ms TTL=128

Ping statistics for 172.30.12.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.12.1

Pinging 172.30.12.1 with 32 bytes of data:

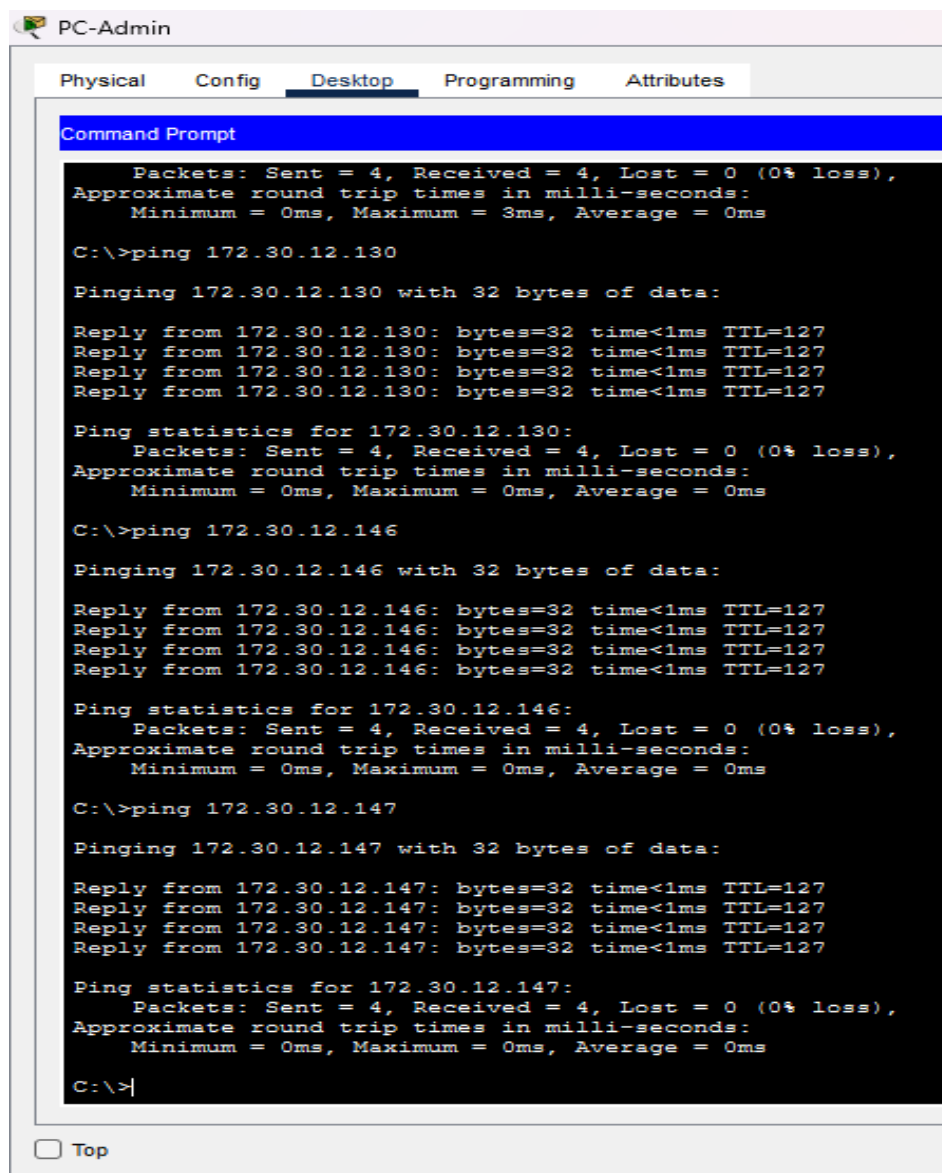
Reply from 172.30.12.1: bytes=32 time<1ms TTL=255
Reply from 172.30.12.1: bytes=32 time<1ms TTL=255
Reply from 172.30.12.1: bytes=32 time=3ms TTL=255
Reply from 172.30.12.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.30.12.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>|
```

At the bottom of the window, there is a "Top" button with a small square icon to its left.

Figure 7.2 — PC-Admin: pings to its own gateway, a same-VLAN host, and another VLAN gateway, 0% loss



The screenshot shows the PC-Admin interface with the Desktop tab selected. A Command Prompt window displays the results of three ping commands. Each command targets a different IP address (172.30.12.130, 172.30.12.146, and 172.30.12.147) and shows a successful response with 0% loss and a TTL of 127. The output for each ping includes the number of packets sent and received, the approximate round trip times in milliseconds, and the TTL value.

```
PC-Admin
Physical Config Desktop Programming Attributes
Command Prompt

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>ping 172.30.12.130

Pinging 172.30.12.130 with 32 bytes of data:

Reply from 172.30.12.130: bytes=32 time<1ms TTL=127
Reply from 172.30.12.130: bytes=32 time<1ms TTL=127
Reply from 172.30.12.130: bytes=32 time<1ms TTL=127
Reply from 172.30.12.130: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.12.130:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.12.146

Pinging 172.30.12.146 with 32 bytes of data:

Reply from 172.30.12.146: bytes=32 time<1ms TTL=127
Reply from 172.30.12.146: bytes=32 time<1ms TTL=127
Reply from 172.30.12.146: bytes=32 time<1ms TTL=127
Reply from 172.30.12.146: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.12.146:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.30.12.147

Pinging 172.30.12.147 with 32 bytes of data:

Reply from 172.30.12.147: bytes=32 time<1ms TTL=127
Reply from 172.30.12.147: bytes=32 time<1ms TTL=127
Reply from 172.30.12.147: bytes=32 time<1ms TTL=127
Reply from 172.30.12.147: bytes=32 time<1ms TTL=127

Ping statistics for 172.30.12.147:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
  Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 7.3 — PC-Admin: inter-VLAN pings to Finance, IT and the server, 0% loss, TTL 127 confirming routing via R1

8. Testing Summary

Test	Expected Result	Result
IPv4 ping: gateway, same-VLAN, inter-VLAN	All reply, 0% loss	PASS
IPv6 ping: gateway, same-VLAN, inter-VLAN	All reply, 0% loss	PASS
SSH login to R1 — correct credentials	Reaches R1#	PASS
SSH login to R1 — incorrect credentials	% Login invalid	PASS
SSH login to SW1 — correct credentials	Reaches SW1#	PASS
SSH login to SW1 — incorrect credentials	% Login invalid	PASS

Telnet to R1 / SW1	Connection refused (SSH only)	PASS
Guest → Guest gateway (IPv4 / IPv6)	Replies normally	PASS
Guest → internal VLANs (IPv4 / IPv6)	Blocked, 100% loss	PASS
Internal PC → SRV1 (HTTP)	Web page loads	PASS
Guest laptop → SRV1 (HTTP)	Request times out	PASS

9. Troubleshooting Notes

There were several problems encountered during the implementation, and these were solved. Here they are documented for transparency and as a record of the troubleshooting process.

- Forgotten device passwords on R1 and SW1 (carried over from an earlier session): both devices were replaced with fresh units and recabled, since no prior configuration needed to be preserved. This also reinforced why the client's "secured administration" requirement matters in practice.
- When trying to log in via SSH, repeated "% Login invalid" errors were encountered because the netadmin password typed during the test did not match the password stored on R1. The solution was to re-set the enable secret and the netadmin password directly on the device console to a known value and then try to log in with SSH again to test that the credentials were the same.
- The vty line range on SW1 was initially only configured for lines 0-4 (SSH) and lines 5-15 were left open with no login configured and were subsequently locked down using the same login local and transport input ssh settings, leaving an otherwise unused gap.
- All configuration changes were saved with copy running-config startup-config on both R1 and SW1 after each change, and verified to persist.

10. GitHub Portfolio

All project evidence, configuration exports, the Packet Tracer file, and this documentation are maintained in the GitHub repository below.

https://github.com/blessedmaake18-bee/Tswalu_Network

Folder	Contents
configurations/	Router_R1_Config.txt and Switch_SW1_Config.txt (full running-configs)
diagrams/	Network topology diagrams
docs/	Design_Document.pdf, Network_requirements.pdf, this implementation report
ip-addressing/	IPv4/IPv6 addressing plan
packet-tracker/	Tswalu_Network.pkt — working Cisco Packet Tracer file
screenshots/	All configuration and testing evidence referenced in this document

11. Conclusion

The goal of this project was to take the original network design, from a paper proposal to a working, fully tested implementation, using Cisco Packet Tracer. All of the requirements were discussed intentionally, not just by default: router-on-a-stick was selected for all five VLANs were routed through a single link to R1, but still had native, independently verifiable IPv4 and IPv6 on each segment; and guest isolation was added using ACLs on R1 itself, the one point from which all inter-VLAN traffic must flow out from, not with a separate appliance. For each step of moving through the various configurations, an informational test was performed that would fail if it was incorrect; no configuration step was performed with a test that could only confirm success. Each step of the various configurations was accompanied by an informational test that would fail if it were performed incorrectly, but never a test that would only pass if it was correct.

Implementations also came up and addressed real problems that were the concern of the client: forgotten passwords on both devices, a credential mismatch during testing on SSH, and an unsecured VTY range on SW1 that was detected prior to final testing (Section 9). Addressing these appropriately, instead of avoiding them, made it relevant to understand the importance of secured administration in practice. As implemented, the network of Tswalu Steel Fabrication, satisfies all stated requirements and the ability to explain, defend and extend the solution as evidenced through CLI, ping and browser testing, and a documented troubleshooting trail, gives confidence that the solution is well understood.